**R16**

**Code No: 133BC**

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**
**B.Tech II Year I Semester Examinations, October - 2020**
**MATHEMATICAL FOUNDATIONS OF COMPUTER SCIENCE**
**(Common to CSE, IT)**

**Time: 2 hours** **Max. Marks: 75**

**Answer any five questions**
**All questions carry equal marks**
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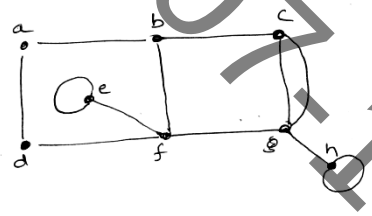
1.a) Obtain principal conjunctive normal form (PCNF) for the formula $(\sim p \rightarrow r) \wedge (q \leftrightarrow p)$
b) Show that the following premises are inconsistent $P \rightarrow Q$, $R \rightarrow S$, PVR, ~(QVS). [8+7]

2.a) On the set Q of all rational numbers, the operation * is defined by a*b=a+b-ab. Verify that under this operation, Q forms a group or not?
b) Let A = {1,2,3,4}, then write all possible subsets of A. [8+7]

3.a) Write the 3- combinations and 3-permutations of {3.a, 2.b, 1.c, 3.d}.
b) Find the number of non negative integral solution for the equation $X_1+X_2+X_3=40$, where $X_1 \geq 1$, $X_2 \geq 5$, $X_3 \geq -4$, $X_4 \geq 3$. [7+8]

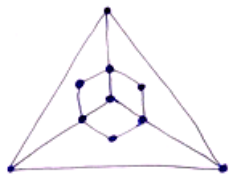
4.a) Find the solution for the Fibonacci series $a_n=a_{n-1}+a_{n-2}$, $n \geq 2$ and $a_0=1$, $a_1=1$.
b) Find the solution for $a_n-3a_{n-1}-4a_{n-2}=6^n$ for $n \geq 2$ and, $a_0=a_1=1$. [8+7]

5.a) Find the dual of the following graph 1.




Graph: 1

b) Find the chromatic number for the following graph 2. [7+8]



Graph: 2

6.a) Using automatic theorem proving, prove that $(\sim Q \wedge (P \rightarrow Q)) \Rightarrow \sim P$.
b) Obtain principal disjunctive normal form (PDNF) for $p \rightarrow ((p \rightarrow q) \wedge \sim(\sim q \vee \sim p))$. [7+8]

7.a) Verify the system G = ({2, 4, 6, 8}, $X_{10}$) where $X_{10}$ is "multiplication module 10" operation is group or not?
b) Let A = {1,2,3,4,5,6,7}, B = {4,5,6,7,8,9} then find:
i) $A \cup B$ ii) A - (A-B) iii) (A-B) $\cup$ (B-A) iv) $(A \cap B)$ v) A-$(A \cap B)$. [7+8]

8. Find the number of ways of distributing 28 similar balls into 5 numbered boxes where each box contains at least 3 balls. [8+7]

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JNTUH USED 07-11-2020 AM